Arista Access Node IND-ACC-16 Optical Transport Failure SOP (ALARM_3088)

Node: IND-ACC-16 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_3088

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the Access Node IND-ACC-16 can be attributed to the inherent limitations of the Optical Transport Network (OTN) protocol, coupled with the specific hardware implementation on the Arista equipment. The OTN protocol relies heavily on the transmission and reception of optical signals over fiber optic cables, which can be susceptible to degradation due to various environmental factors, such as temperature fluctuations, humidity, and physical damage. The Arista equipment, being an OTN-enabled device, is designed to monitor and report on the health of these optical signals. However, in the event of a failure, the equipment may not always accurately report the root cause, leading to potential misdiagnosis. Additionally, the Access Node's physical location in the Indiranagar Exchange, which is prone to high traffic and environmental stressors, further exacerbates the likelihood of optical transport failures. Therefore, it is essential to carefully diagnose and troubleshoot these failures to ensure timely resolution and prevent potential service disruptions.

Pre-requisite Checks

- Verify that the Indiranagar Exchange has been dispatched with a remote hands team to assist with physical access and environmental checks.

- Check the redundant link status to ensure that an alternate path is available in case of a failure.

- Confirm that the Arista equipment is running the latest software version and that all necessary patches have been applied.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to high temperature or humidity	High
Fiber cuts or physical damage	Medium
Rx/Tx power loss due to optical signal attenuation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface Status: Online, Link is up, Description: This command verifies the status of the optical interface and confirms whether the link is up or down. A down link may indicate a physical issue or a software problem.

Command: show optic transceiver detail, Expected Output: Transceiver Details: Temperature: 45.6C, Optical Power: -3.2 dBm, Description:
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```
This command provides detailed information about the transceiver, including temperature and optical power levels. High temperature or low optical power may indicate a problem with the transceiver or the fiber optic cable.

Command: show optic transceiver alarm, Expected Output: Alarm Status: Transceiver alarm is active, Description: This command checks the alarm status of the transceiver and confirms whether an alarm is active. An active alarm may indicate a problem with the transceiver or the fiber optic cable.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optic transceiver {transceiver-slot}, Description: This step involves replacing the SFP/XFP module with a new one. The remote hands team should physically replace the module and verify that the link comes back up.

Step 2: Step: Verify the fiber optic cable, Command: verify optic transceiver cable {transceiver-slot}, Description: This step involves verifying the integrity of the fiber optic cable. The remote hands team should visually inspect the cable for any signs of physical damage or cuts.

Step 3: Step: Adjust the Rx/Tx power levels, Command: set optic transceiver power {transceiver-slot} {power-value}, Description: This step involves adjusting the Rx/Tx power levels to compensate for signal attenuation. The remote hands team should use a power meter to measure the optical power levels and adjust the settings accordingly.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface Status: Online, Link is up, Description: This command verifies the status of the optical interface and confirms whether the link is up or down.

Command: show optic transceiver detail, Expected Output: Transceiver Details: Temperature: 25.6C, Optical Power: -2.2 dBm, Description: This command provides detailed information about the transceiver, including temperature and optical power levels. Normal operating temperatures and optical power levels should be within the specified ranges.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: switchover optic transceiver {transceiver-slot}, Description: This step involves restoring the original SFP/XFP module. The remote hands team should physically restore the original module and verify that the link comes back up.

- Step: Verify the fiber optic cable, Command: verify optic transceiver cable {transceiver-slot}, Description: This step involves verifying the integrity of the fiber optic cable. The remote hands team should visually inspect the cable for any signs of physical damage or cuts.